

AURA — Complete User Instruction Manual

Firmware: AURA 1.0.0 · Mark III "Phoenix" · Development channel

Platform: ESP32-WROOM-32 · Arduino-ESP32 core

Companion app: AURA Companion 1.1.0 (Android / Windows)

Document version: 1.0 — generated from the current codebase (source of truth)

How to Read This Manual

This manual describes exactly what the **current** AURA firmware and companion app actually do, verified from the source code. Three markers are used throughout:

Marker	Meaning
Ready	Implemented and working in this build
Requires configuration	Code exists but needs a credential, TLS certificate, or hardware to produce results
Not currently available	Code path exists but is not wired up, or the hardware / API is absent

Source-of-truth rule. Where this manual conflicts with any earlier README or marketing material, the *current code* wins.

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1. What AURA Is

AURA is a small, personal, **voice-first AI assistant** built on a single low-cost ESP32 microcontroller. It sits on your desk, shows its mood through a miniature OLED "face" and a 16-LED ring, listens through a microphone, and answers through a speaker.

AURA handles the entire loop on one chip:

- **Hear** — an INMP441 I2S MEMS microphone captures your voice.
- **Think** — speech-to-text turns your words into text; a reasoning engine produces an answer. When the network is unavailable, a fully **offline Local AI Engine** answers instead.
- **Speak** — text-to-speech converts the answer into audio for an amplifier and speaker.
- **Show** — a 128×64 monochrome OLED renders the idle "face", a clock, and every state; the 16-LED ring uses a single distinct solid colour per state to communicate the same state at a glance.
- **Connect** — Wi-Fi, an embedded web configuration portal, and the AURA Companion app for phone and PC.

AURA also includes a device-side productivity suite: reminders, goals, habits, a planner, study tools, daily briefings, a knowledge graph, documents, workspaces, semantic search, decisions, predictions, and recommendations. These store their data on the ESP32's on-chip filesystem and are managed through the web portal or

the companion app.

Platform note. AURA runs on a very small microcontroller (≈ 3 MB app flash, ≈ 320 KB RAM, no PSRAM). That constraint, plus hardware and API availability, is why some features in this manual are marked *Requires configuration* or **Not currently available**.

2. Capability Summary

Capability	Status	Notes
Touch interaction (single tap, double tap, 5 s hold)	Ready	See §5
Microphone capture (INMP441, I2S, 16 kHz)	Ready	See §6
Voice wake word (always-listening mic)	Not currently available	Wake-word code exists but is not wired; touch (push-to-talk) is the only input
Speech-to-Text (Sarvam AI, cloud)	Requires configuration	Needs a Sarvam API key and TLS root CA in <code>secrets.h</code> (empty by default)
Text-to-Speech (Sarvam AI, cloud)	Requires configuration	Same requirements as STT
Speaker / spoken reply (MAX98357A)	Hardware-dependent	Pins are defined, but the shipped build is compiled with the speaker not present
Gemini reasoning (cloud)	Not currently available	Model constant exists but the API endpoint is not wired in this build; the offline engine is used
Offline Local AI Engine	Ready	Fully offline, always available
Startup greeting	Ready	Configurable, ~ 4 s display
OLED display (SH1106, 128 $\times$ 64)	Ready	Face, clock, states
LED status ring (WS2812B, 16 LEDs)	Ready	See §8
Disco Mode (party lights)	Ready (app-only)	Fully separate from status; see §8
Privacy / mic mute	Ready	App + OLED + magenta ring
Wi-Fi STA + soft-AP + captive portal	Ready	See §14
NTP time sync	Ready	Powers the idle clock
Web configuration portal	Ready	See §14, §23
Embedded REST API (100+ endpoints)	Ready	See §23
OTA firmware updates	Ready	Web upload; see §17
microSD card storage	Requires hardware	Card is optional; on-chip SPIFFS is the default store

Sensors (BME280, BH1750, IMU, GPS, etc.)	Not currently available	No sensor modules exist in this firmware build
Battery gauge	Not currently available	No battery-sense code; the app shows a placeholder
Companion app	Ready	See §12

3. Hardware Overview

The table lists the hardware actually supported by the current firmware, with pins verified from `config.h`.

Component	Function	Connection / pins	User-visible purpose
ESP32-WROOM-32 (38-pin)	Main MCU (Wi-Fi + BLE)	—	All processing and wireless
SH1106 OLED, 128×64 px	Display	I2C SDA=GPIO21, SCL=GPIO22, addr 0x3C	Face, clock, states
INMP441 I2S MEMS microphone	Voice input	BCLK=GPIO26, WS=GPIO25, DATA=GPIO34	Captures your speech
MAX98357A I2S amp + speaker	Voice output	BCLK=GPIO27, LRC=GPIO14, DATA=GPIO12	Speaks replies (hardware-dependent)
WS2812B 16-LED ring	Status lights	WS2812 on GPIO4	Solid status colours
TTP223 capacitive touch	Input	GPIO13, active-high	Tap / double-tap / hold
microSD / SD module	Optional storage	CS=GPIO5, MOSI=GPIO23, MISO=GPIO19, SCK=GPIO18	Optional data and asset storage
Wi-Fi + Bluetooth (integrated)	Networking	onboard	Connection, app, cloud

Speaker note. The firmware is compiled with `AURA_HW_SPEAKER_PRESENT = 0`.

The I2S speaker stage is disabled and boot logs "Speaker output skipped (not connected)". On this unit you will **not** hear spoken replies until a MAX98357A (or equivalent) is connected and the build flag enabled. Microphone, OLED, LED ring, touch, Wi-Fi, and storage are unaffected.

Not in this firmware (parts-list only): a 5" touchscreen, GPS, NFC, camera, RTC module, IMU, distance sensor, ambient-light sensor, temperature / humidity / pressure sensor. None of these have firmware support in the current build and are not documented here as working functions.

4. Power-On, Startup, Restart, Shut Down

4.1 What happens when you apply power

1. **Connect 5 V / USB-C power** (or a battery, if fitted).

1. The ESP32 boots; a serial console at **115 200 baud** shows progress if a computer is connected.

1. **System Manager** initialises the ~57 modules in dependency order — storage, display, Wi-Fi, UI framework, audio, LED ring, AuraSystem, conversation, memory, planners, OTA. This typically finishes in a few seconds.

1. **OLED:** boot screen — a small logo with a pulsing ring, the text "AURA", a progress bar, then **READY**.

1. **LED ring:** solid blue during boot, then the **idle** indication (solid blue with a subtle flicker).

1. **Network:** if saved Wi-Fi credentials exist, AURA joins that network (STA); otherwise it starts a soft-AP named `AURA_Setup` so you can provision it. On success it syncs time from NTP.

1. **READY.** The OLED shows the idle face (and clock once time is synced); the ring shows the solid-blue idle light.

1. Optional **startup greeting** is displayed for about 4 seconds (unless disabled).

4.2 What you should see

The OLED completes the boot animation and shows the face; the ring shows the idle solid-blue light; the unit responds to a single tap. No permanent error text, no reboot loops.

4.3 If it does not boot

- Check a stable **5 V** supply and a good USB-C cable.
- Unplug power, wait ~5 seconds, reconnect.
- Watch the serial console. Repeated `rst:` lines or "Guru Meditation" indicate a crash loop. The firmware counts failed boots and then enters **Safe Mode** — a reduced load state that lets you reach the web portal, review crash logs, and recover.

- If serial boots fine but the OLED stays blank, check OLED wiring / I2C address

(§21).

4.4 Restart and power off

ESP32 firmware has no software power-off; to power off simply remove power. To restart without unplugging:

- **Web portal:** Restart button (or `POST /api/restart`).
- **Companion app:** Control → Restart device.

Factory reset wipes device storage (settings, Wi-Fi, memories, reminders, goals, habits, secrets) back to defaults. Use `POST /api/factory-reset` or the app Control → Factory reset (both confirm first). After a reset you must re-provision Wi-Fi and sign in again.

Warnings

- Never remove power during an OTA update (§17).
- A factory reset is **irreversible**.

5. Touch Sensor — Complete Guide

The touch pad (TTP223, active-high) is the only physical input. AURA polls it every ~20 ms and applies debounce and timing so taps, double taps, and holds are distinguishable.

5.1 Gesture grammar

Gesture	Timing	Result
Single tap	pressed ~50–400 ms, released; no second tap within 400 ms	Start a conversation — microphone ON (listening)
Double tap	two taps, second within 400 ms	Cancel / stop the active interaction → return to idle
5-second hold	press held 5000 ms without release	Enter Setup Mode (§14) — highest priority
Any other hold (>400 ms but <5 s)	press then release	Ignored — deliberately nothing

Priority on a single press: **5 s hold** > **double tap** > **single tap**.

5.2 Behaviour in context

- **Idle:** single tap opens the mic and starts a conversation.
- **During listening / recording / processing / speaking:** single tap is ignored;
double tap cancels the interaction and returns to IDLE.
- **Privacy / muted:** touch **never** opens the mic; the display shows "Microphone Muted".
- **Setup mode:** gestures are disabled except a further 5 s hold, which exits setup.
- **Auto-sleep:** any touch wakes AURA; a 5 s hold goes straight into setup.

In short, "**tap once to talk, tap twice to stop, hold five seconds for setup**".

There is no wake-word.

6. Microphone Guide

6.1 What the microphone is

The INMP441 digital microphone (I2S, 16 kHz, 16-bit mono) captures your voice so it can be transcribed. By default it is **push-to-talk**:

- **MIC INITIALIZED** — the driver is created and pins configured at boot.
- **MIC ACTIVELY LISTENING / RECORDING** — only while a conversation is active (after a single tap).

The mic is **not always-on**. At boot the firmware stops recording and does not enable voice-activity signals, so no ambient sound is captured until you tap.

6.2 How to activate it

Single tap the touch sensor. AURA opens the mic and the OLED/LED switch to the listening state.

6.3 How to stop it

- **Double tap** anywhere stops the interaction and returns to IDLE.
- Otherwise the capture stops automatically once you stop speaking (silence end), then STT → AI → TTS complete the turn.

6.4 Knowing what the mic is doing

Signal	Meaning
OLED label "Listening..."	Mic is capturing
LED ring solid cyan (then green while recording)	Mic is recording
Mic icon active	Mic active
Mic icon muted / OLED "Microphone Muted"	Mic is muted
LED solid magenta	Muted / privacy

6.5 Privacy / mute

- Mute is controlled from the companion app or the web portal.
- While muted: recording is prevented, voice-activity detection is off, no audio is sent; the OLED shows the muted notice and the ring is solid magenta.
- Unmute restores voice behaviour.

6.6 Wake word — Not currently available

A wake-word handler exists in the conversation manager but nothing invokes it in this build. **Push-to-talk via touch is the only input path.** Wake-word is documented as not available.

6.7 If STT is unavailable

If speech-to-text cannot initialise (missing key/TLS or no network), the voice turn degrades gracefully. **This is not a sign that the touch sensor is broken** — always check the LED and OLED to separate a touch failure from an STT failure (§21).

7. OLED Display Guide

The display is a **128×64 monochrome SH1106**. Colour is conveyed through the LED ring, not the screen.

7.1 User-visible states

Screen	Meaning	What to do
Boot animation + "AURA" + progress + "READY"	Starting up	Wait
Idle face + optional clock	Ready and idle	Tap once

"Listening..."	Mic is capturing	Speak
"Thinking..."	AI is processing	Wait
"Speaking..."	Playback running	Wait (needs speaker)
"AURA SETUP" + SSID	Setup / provisioning	Join the SSID (§14)
"Microphone Muted"	Mute active	Unmute from the app
Error / notice screen	A failure state	See §21
OTA screen	Updating	Keep power

7.2 Mic icon and idle clock

- **Mic icon** (top right) reflects mic mode: idle, listening, or muted.
- **Idle clock**: while the face is genuinely idle, the top shows an `HH:MM` clock

in **India Standard Time (IST, UTC+5:30)**. It appears only when the face expression is `IDLE` ; any active state replaces it. Until NTP sync, it shows `--:--` .

8. LED Ring Guide (Status and Disco are separate)

There are **two independent LED systems**: the normal status indicator and Disco Mode. Keep them separate.

8.1 Normal status LED (FINAL: solid colour per state)

When not in Disco mode, the 16-LED ring is **ONE indicator**: every AURA state paints **ALL 16 LEDs in a single, distinct SOLID colour**. There is **no** LED travel, chase, comet, tail, pulse, breathing, or rotation in the status path.

The only variation allowed is a subtle, ****whole-ring synchronised brightness flicker**** — all 16 LEDs share one random-walk brightness level (~85–100 % for normal states, ~92–100 % for quiet states), recomputed every 50–150 ms. It looks like gentle electronic "energy", not a flashing alarm. The error state uses a deeper, faster flicker so it stands out.

State	Solid colour
IDLE / READY	Solid Blue <code>#0080FF</code>
BOOT	Solid Blue <code>#0080FF</code> (matches IDLE identity)
LISTENING	Solid Cyan <code>#00FFFF</code>

RECORDING	Solid Green #00FF40
THINKING / PROCESSING	Solid Yellow #FFFF00
SPEAKING	Solid White #FFFFFF
SETUP	Solid Purple #8000FF
PRIVACY / Muted	Solid Magenta #FF00AA
OTA	Solid Orange #FF8000
Wi-Fi connecting	Solid Steel blue #4080E0
Wi-Fi connected	Solid Spring green #00FF80 (transient → IDLE)
Happy	Solid Gold #FFD700
Success	Solid Mint green #00FFAA (transient → IDLE)
Reminder / notification	Solid Amber #FFAA00
Warning	Solid Red-orange #FF4000 (error-style flicker)
Critical	Solid Dark red #8A0000
Offline	Solid Slate #607085
Sleep / Wake	Solid Dim navy #404A60 / Azure #00AFFF

Brightness is adjustable (default ~120). IDLE glows a quiet dim-blue presence instead of turning off. Colour changes cross-fade crisply between SOLID states.

Privacy / muted states use solid **magenta** (not red).

8.2 Disco Mode (separate, app-only)

Disco Mode is a fun, optional **visual override**, turned on from the Companion app (Tools → Disco Mode) or the portal API.

- Turns the ring into **party lights**: a rotating set of ~10 smooth colours (deliberately low-strobe, eye-friendly) at a configurable brightness.
- **Overrides** the normal status animation while enabled.
- **Emergency exception**: a critical error temporarily breaks through with a red alarm, then Disco resumes.
- **Not persisted**: Disco is OFF after every boot; enable it again from the app.
- Disco leaves the OLED and the microphone untouched.

Disco is not the normal status indicator and never claims to be.

9. Voice Pipeline

User → Touch (tap) → Microphone (INMP441) → Voice capture
 → PT (Sarvam, cloud) → Reasoning (Local AI by default; Gemini if configured)
 → TTS (Sarvam, cloud) → Speaker (MAX98357A, hardware-dependent)

Stage	Component	Internet required?	Current state
Input	TTP223 touch	No	Ready
Capture	INMP441 I2S mix	No	Ready
STT	Sarvam STT (cloud)	Yes	Requires configuration
Reasoning	Local AI Engine (offline)	No	Ready (default)
Reasoning (optional)	Gemini (cloud)	Yes	Not currently available
TTS	Alternative TTS (cloud)	Yes	Requires configuration
Output	MAX98357A	No	Hardware-dependent

What this means today: you can always interact by touch, and the assistant can always answer thanks to the offline engine. For the audible voice conversation you must supply the speech configuration and the speaker hardware; voice-output pipes are included in the code, but the shipped build has the speaker off.

10. Sarvam Speech-to-Text / Text-to-Speech

10.1 What they do

- **STT (Speech-to-Text):** streams the microphone capture (16 kHz WAV) to the Sarvam API over HTTPS and returns a transcript. Voice-activity detection determines where your speech ends.

- **TTS (Text-to-Speech):** takes the assistant's answer, requests Sarvam, and streams audio back (Base64 → PCM at 16 kHz) for playback.

Both are **cloud services**: they require an Internet connection and a valid account key.

10.2 Configuration

Secrets are never committed. A template exists at `secrets.h.example` :

```
copy secrets.h.example secrets.h
# in secrets.h set:
```

```
# SARVAM_API_KEY = "..." (Sarvam account key)
# SARVAM_ROOT_CA = "..." (TLS root certificate of the API host)
```

- Keys load from NVS at runtime; empty values make initialisation fail closed.
- **TLS is strict:** without a root CA certificate, STT/TTS cannot initialise.
- The language model expects Indian English and Hindi support (see `config.h`).

10.3 Failure behaviour

If STT or TTS cannot start (missing key/CA, offline), that stage is skipped and reported clearly. **An STT failure is never a fault of the touch sensor.**

Note: some architecture notes call these modules "placeholders"; in the current source they are **fully implemented** (WAV upload, TLS, streaming decode). They are gated purely on configuration and network.

11. Reasoning — Gemini (Cloud) and the Local AI Engine

11.1 Gemini (cloud)

- Requested model: `gemini-3.5-flash-lite` at Google's `generateContent` endpoint (see `config.h`).
- Needs a Google API key, TLS root CA, and Wi-Fi; uses a timeout and a circuit breaker.
- On any failure (no Wi-Fi, no key, no endpoint, no CA), callers fall back to the offline engine.

Setup status: the model string is defined, but the API endpoint is ****never** wired into the client** in this build (`GeminiClient::setApiEndpoint` has no callers). Gemini is therefore **Not currently available**, and the assistant uses the Local AI Engine. This is offline-first behaviour by design.

11.2 Local AI Engine (offline)

The offline engine builds replies entirely on-device, using conversation context, memories, goals, the planner, personality, time of day, and sentence generation

with a response cache. It is always available, works without any network or API

key, and can be enabled/disabled from the portal (`/api/offline_ai/*`).

12. The AURA Companion App

The companion app is a Flutter application for **Android** and **Windows**.

12.1 Install and first launch

1. Install / build the app, then launch **AURA Companion**.

1. A splash screen leads to a **sign-in card**, which depends on the connection

mode:

- **Local device reached** → device sign-in (username + password). The admin

password is printed on the serial monitor at first boot (development builds

prefill `Devil` / `Devil` for local testing only).

- **Device unreachable** → a "device unavailable" message with a path to

cloud sign-in.

- **Remote (cloud) mode** → email/password sign-in with Sign Up and Forgot

Password, via Supabase.

12.2 Home navigation

The app shows the main destinations as a rail (wide screens) or bottom bar

(phones):

Tab	Purpose
Dashboard	Live device overview — connection badge, module status, resource gauges (CPU, storage, temperature, battery), metric grid (heap, Wi-Fi, uptime)
Chat	Text chat with the on-device assistant; replies are memory-aware
Tools	Launchpad for device tools (\$19)
Alerts	Local notification settings and a test-alert button
Settings	App preferences, speech settings, and (hidden) developer settings

12.3 Connecting the app to the device

- Default addresses: REST `http://:80` , WebSocket `ws://:81` ; AP fallback

address `192.168.4.1` ; mDNS hostname `aura-.local` .

- The app automatically probes the device, validates the stored session, and

routes you to the dashboard when possible.

- Networking details are intentionally hidden behind the **developer settings** screen.

12.4 Notifications

Alerts and reminders use **local notifications** inside the app (no cloud push service). The Alerts screen offers: device alerts on/off, alert sound/vibration, and reminder notifications. A **Test alert** button lets you verify delivery.

13. Local vs Remote Mode

	Local (device)	Remote (cloud)
What it is	App talks directly to the device	App talks via Supabase
When	Device reachable on the LAN	Device out of range
Requirements	Same network, device powered	Supabase project + account
If unreachable	"Offline" badge	"Offline" / reconnecting
Sign-in	Device username/password	Email/password

- **Local is always preferred.** The app falls back to remote when the direct connection is unavailable.
- The connection badge in the app title shows `Cloud` (remote) or `Connected` (local), plus transient `Connecting` / `Testing` / `Reconnecting` states.
- If the device itself is offline, neither mode can talk to it; the ****Local AI still answers**** on the device, and touch interaction still works.

14. Setup Mode (Wi-Fi Provisioning)

Enter Setup by holding the touch pad for **5 seconds**.

1. **Touch and hold 5 s.** AURA enters Setup Mode.
1. **OLED:** shows **AURA SETUP** plus the SSID to join.
1. **LED:** Solid **Purple** (setup indication; magenta when muted during setup).
1. On your phone, join the Wi-Fi network shown on the OLED. The SSID is `AURA_Setup` (from `Secrets::AP_SSID`); the password is generated at runtime and printed to the serial console unless a placeholder in `secrets.h` is set.

1. Open a browser to **http://192.168.4.1** (or `http://aura-.local`) — the captive portal config page.
1. Enter your home Wi-Fi SSID and password, then save. AURA connects to that network and exits setup.
1. To exit setup manually, hold the pad for 5 seconds again.

While in setup the microphone is disabled and touch input is restricted; this prevents accidental conversations during configuration.

15. Privacy / Microphone Mute

- **Enable mute** from the Companion app (Control) or the Web portal.
- While muted:
 - Mic capture and voice-activity detection stop; no audio is sent.
 - OLED shows **Microphone Muted** and the mic icon shows a muted state.
 - The LED ring becomes **solid magenta**.
 - The touch pad does **not** open the microphone.
- **Unmute** from the app/portal restores voice behaviour (LED returns to idle).

Terminology:

- **MIC OFF** — normal, not currently capturing.
- **MIC MUTED** — intentionally silenced (privacy).
- **MIC INITIALIZED** — driver ready, not capturing (default at boot).

16. Storage and the microSD Card

16.1 On-chip storage (default)

The firmware uses **SPIFFS** (on-chip filesystem, ~896 KB partition) as the primary store. It keeps:

- Conversations (JSON) — auto-promoted to memories
- Memories with revisions
- Reminders, goals, habits, planner data
- Knowledge graph

- Log files (rotating) and crash logs
- Backup / export files

16.2 microSD (optional)

A microSD card (SPI pins above) is optional. If present, it gives extra capacity for data, audio assets, and logs. If **absent**, SPIFFS keeps every feature working — the SD is entirely optional.

- Check status: GET /api/sd/diagnostics in the portal.
- If the card is not mounted, check the SPI wiring and format the card as FAT32.

17. OTA Firmware Updates

- **What:** updating AURA's firmware over Wi-Fi from the app or the web portal, without USB.
- **How:** Portal /ota (browser upload) or app Tools → Firmware → pick a .bin ; live progress is shown.
- **Before you start:** use a stable, mains-powered USB supply. Do **not** unplug during the update.
- **During:** the portal streams the file and verifies SHA-256 (and, when signed, an ECDSA signature) before applying it.
- **Failure:** an invalid image is rejected and the current firmware stays bootable; simply retry.
- **Power loss** during the flash can leave the app invalid — recover by flashing via Arduino IDE over USB serial.

18. Sensors

The firmware does **not** include environmental sensor modules. It does include a **Health Monitor** that samples system metrics:

Metric	What it detects	Alert
Heap (free RAM)	Low memory	Low-heap warning
Wi-Fi RSSI	Weak connection	Weak-signal warning

SD presence	Card mounted?	Missing-card warning
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These are operating-system health metrics (shown in the app dashboard), not temperature/pressure/light sensors. Sensors from the parts list (BME280, BH1750, etc.) are **not currently supported**.

19. Companion App Tools

The **Tools** tab contains:

Tool	Purpose	How to use
Control	Direct device control	Restart, Factory reset, brightness, volume, gain
Disco Mode	Party lights	On/Off toggle + brightness slider
Memory	Browse stored memories	Search, category filter, pin, archive/restore
Reminders	Schedule reminders	Set title/date/time (local notifications)
Firmware (OTA)	Update firmware	Pick <code>.bin</code> , upload, watch progress
SD Card	Browse / upload / delete files	Root path, upload, download, delete
Monitor	System diagnostics	Live CPU/heap/uptime telemetry

Device-control tools (Control, Disco, SD, OTA) are available only when the app is connected in **local** mode; the app hides them when the device is unreachable or the app is in remote mode.

20. Status Reference — Master Table

State	OLED	LED ring	Microphone	App	Meaning
IDLE	Face (+clock when synced)	Solid Blue (subtle flicker)	OFF	Connected	Ready
LISTENING	"Listening..."	Solid Cyan	ON	Mic active	Capturing
RECORDING	"Recording..."	Solid Green	ON	—	Recording for STT
THINKING / PROCESSING	"Thinking..."	Solid Yellow	OFF	Working	AI processing
SPEAKING	"Speaking..."	Solid White	OFF	—	Playback (needs speaker)
MUTED	"Microphone Muted"	Solid Magenta	OFF (muted)	Muted	Privacy
SETUP	"AURA SETUP" + SSID	Solid Purple	OFF (blocked)	—	Provisioning
ERROR	Error / notice	Solid Red (deep fast flicker)	—	Error	Failure

OTA	Update screen	Solid Orange	—	Updating	Firmware update
DISCO (extra)	Unchanged	Party animation	—	Disco toggle	Visual override

App-only entries: **Disco** (controlled only from the app/portal; OLED unchanged).

21. Troubleshooting

Symptom	Likely causes	Check	Solution
AURA won't boot	Power/bad cable, crash loop	Serial output, LED behaviour	Stable 5 V; re-flash over USB; Safe Mode appears after repeated crashes
OLED blank	Display wiring / I2C address / power	Address 0x3C, serial logs	Reconnect/verify address; re-flash if needed
OLED animations stop	Watchdog / hang	Serial log	Reboot; re-flash if persistent
LED ring off	LED power rail, GPIO4 wiring, brightness 0	Solid boot blue	Power the ring correctly; verify pin
Touch not responding	Sensor wiring / ground / muted state	Try repeat taps; watch LED	Check wiring; not related to STT
Mic never starts	Continuous-listening setting, muted	App settings, LED magenta	Keep push-to-talk; unmute; single tap
Mic won't activate	Privacy/mute active	LED magenta, OLED muted	Unmute from app; retry tap
Mic has no audio	Speaker not connected (hardware)	—	Expected until a speaker is fitted
STT unavailable	Key/CA not set, no network	Serial "STT init failed"	Configure Sarvam key + root CA; not a touch fault
TTS unavailable	Same as STT + speaker	—	Configure key/CA; add speaker
Gemini unavailable	Key/endpoint/network	—	Expected now; local engine answers
Wi-Fi unavailable	Power / interference	<code>/api/wifi</code>	Re-scan and reconnect in the portal
App cannot connect	IP changed / different network	Developer → address	Update IP; same network/VLAN
App shows Offline	Device offline or remote mode	Badge state	Reconnect local; re-check Wi-Fi
Cloud login fails	Supabase not configured	<code>.env</code> , supabase docs	Configure project; correct account
SD card not seen	Format/wiring/power	<code>/api/sd/diagnostics</code>	Reformat FAT32, verify SPI pins
OTA fails	Network / wrong file / signature	Hash check	Stable power; retry; signed firmware
Setup opens by itself	Accidental 5 s hold	Press timing	Hold 5 s again to exit
Restarts / watchdog loop	Crash loop, low memory	Serial + crash log	Safe Mode, disable modules, re-flash

Low memory	Too many features	Heap metric	Disable unused modules; free heap
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22. Safety

Power and charging

- Use a quality **5 V / 2 A** USB-C supply. Do not exceed rated current.
- The parts list uses a **5000 mAh Li-ion cell**. Charge it only through a dedicated charger (BQ24074-class). Never charge a bare cell directly.
- Keep the cell inside the enclosure and protect against over-charge, over-discharge, and high temperature.

Wiring and heat

- The ESP32 and WS2812 ring can get warm; keep the enclosure ventilated and away from fabrics/plastics.
- Wire only per the verified pinout in §3; avoid loose strands shorting pins.
- The MAX98357A can be loud — use an appropriate speaker and volume.

Firmware

- Never remove power during an OTA update.
- Prefer a stable serial connection when re-flashing.

Liquids and shorts

- Keep AURA away from all liquids. If wet, disconnect power immediately, dry fully, and inspect before reuse.
- Never insert metal objects into the USB port or touch test pads.

23. Developer / Advanced Section

Architecture

A single Arduino (ESP32) sketch built from ~57 `*Manager` modules, orchestrated by `SystemManager` in dependency order. A 30-second watchdog plus per-module boot progress keeps the system observable and self-recovering (Safe Mode after repeated crashes).

- `config.h` — identity, pins, model strings, hardware flags
- (`AURA_HW_SPEAKER_PRESENT` , `AURA_DEVELOPMENT_MODE`), timing constants.
- `secrets.h` — all keys/credentials; **gitignored; never commit.**
- `version.h` — semantic version (1.0.0 · Mark III "Phoenix").

REST / WebSocket layer

- `web_portal.cpp` — `ESPAsyncWebServer` on port **80** + WebSocket on port **81**;
- ~100 API routes: `/api/status` , `/api/dashboard/` , `/api/memories/` ,
`/api/goals` , `/api/habits` , `/api/planner/` , `/api/reminders/` ,
`/api/skills*` , `/api/automations` , `/api/documents/` , `/api/workspaces` ,
`/api/plugins` , `/api/espnow/` , `/api/companion/` , `/api/diagnostics` ,
`/api/offline_ai/*` , and more.
- REST requires the `X-Auth-Token` header; WebSocket `/ws` carries live status and assistant responses.
- Static UI files live in `data/` (uploaded to SPIFFS).

Storage

- SPIFFS is the default store; `StorageManager` exposes the memory/conversation/audio/reminder/backup/log paths with rotation.

Voice pipeline internals

- `ConversationManager` states: IDLE → LISTENING → PROCESSING → SPEAKING → COMPLETED / ERROR, with interruption and cancel.
- STT/TTS go through `SpeechToTextProvider` / `TextToSpeechProvider` interfaces; Sarvam is the implemented provider.

Constraints

- ~4 MB flash (3 MB app + ~896 KB SPIFFS), no PSRAM (~320 KB RAM). Memory is intentionally tight; large optional features can be disabled to free heap.
- Single OTA app slot limits update size to ~3 MB.

24. FAQ

1. **How do I wake AURA?** Single tap. Voice wake is not available.
1. **How do I turn the microphone off?** Double tap, or mute from the app.
1. **How do I know AURA is listening?** OLED "Listening...", LED solid cyan, mic

icon active.

1. **What does the green LED mean?** Recording state (solid green). Solid cyan is listening, solid yellow is thinking, solid white is speaking.
1. **What does a solid magenta LED mean?** Muted / privacy. Solid red means an error (or warning).

1. **How do I enter setup?** Hold the touch pad for 5 seconds.
1. **Does AURA need Wi-Fi?** For cloud features and NTP yes; the local AI works offline.

1. **Can AURA work without the cloud?** Yes — offline Local AI engine.
1. **What happens if STT fails?** It fails closed; the touch sensor is not broken.
1. **How do I use Disco Mode?** App → Tools → Disco Mode.
1. **How do I update AURA?** App → Tools → Firmware, or the portal `/ota`.
1. **How do I connect the app?** Same network + device login, or cloud sign-in.
1. **Why is the microphone not responding?** Check mute/privacy (LED magenta) and try a single tap.

1. **Why is the OLED blank?** Check wiring/address; use the serial console.
1. **What does the mic icon mean?** Idle, listening, or muted.
1. **My Wi-Fi changed — what do I do?** Re-enter via setup (5 s hold).
1. **Where is my data stored?** On-chip SPIFFS; SD if mounted.
1. **What does factory reset do?** Erases settings, memories, Wi-Fi.
1. **Are notifications real-time?** No — local notifications while the app runs.
1. **Do I need the SD card?** No — SPIFFS always works; SD is optional.
1. **What is Safe Mode?** Recovery state after repeated crashes.
1. **When do I see the clock?** When idle and NTP-synced (IST).
1. **Can I control LED brightness?** Yes — Control + settings.
1. **Can I read logs?** Portal log viewer / developer export.
1. **Is the mic always on?** No — push-to-talk by touch is the default.

End of document.